

Phillip Hirsch

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PROFESSIONAL SUMMARY

Software engineer with over three years of professional experience building web and mobile applications in React, React Native, and TypeScript, shipping production features end to end across mobile and web in a regulated banking environment. Daily fluency with modern AI coding tools, building toward fullstack AI engineering that combines frontend depth with growing backend and agent-focused product work.

WORK EXPERIENCE

Brooksource (Contract with Ally Financial)

March 2023 – Present

Software Engineer

Charlotte, NC

- Built the Welcome experience for Ally's mobile banking app, an Instagram-stories-style onboarding carousel, using XState for state-machine modeling, later refactored to Zustand.
- Built the Explore Products and Rates feature, shipped to both the mobile banking app and a microfrontend on Ally's website, with TanStack React Query handling Contentful data for cross-platform reach.
- Integrated LaunchDarkly to feature-flag both features across mobile and web surfaces, enabling progressive rollout and instant revert capabilities.
- Bridged the JavaScript and native layers of Ally's brownfield React Native app with targeted Swift and Kotlin module changes, enabling blocked features to ship.
- Developed and maintained reusable UI components in React, React Native, and TypeScript that standardized the UI across a number of enterprise applications.
- Improved accessibility compliance (WCAG 2.1) on customer-facing surfaces.

PROJECTS

Personal Portfolio

philliphirsch.com

- Designed and built a personal portfolio site with Astro 6, React 19, and Tailwind CSS v4, deployed on Vercel as a hybrid of static pages and a serverless function.
- Structured the UI with an atomic design component system (atoms, molecules, organisms) and an auto-generated Open Graph endpoint via @vercel/og.
- Enforced code quality with TypeScript in strict mode, ESLint, and Prettier, running on the Bun runtime.

AI-Powered Text-to-Speech for the Visually Impaired

University of North Carolina at Charlotte

- Designed and implemented a deep learning model using PyTorch and fastai, achieving 95% text recognition accuracy to assist visually impaired users.
- Developed a deep learning-based Optical Character Recognition pipeline using convolutional neural networks to recognize and convert over 85% of handwritten text into speech.

EDUCATION

University of North Carolina at Charlotte, Master of Science in Computer Science

December 2022

University of North Carolina at Charlotte, Bachelor of Science in Computer Science

May 2021

SKILLS

Languages: TypeScript, JavaScript, Python, Swift, Kotlin

Libraries & Frameworks: React, React Native, Astro, Tailwind CSS, Zustand, XState, TanStack React Query, React Router, React Navigation, React Reanimated

APIs & Data: REST APIs, GraphQL, Contentful (headless CMS) **Testing:** Jest, Cypress

DevOps & Infrastructure: GitHub Actions, Docker, Vercel, Netlify, Cloudflare, Vite, LaunchDarkly, Dynatrace

AI Tools: Claude Code, Codex, Cursor, OpenCode, Pi